Exercise 2.10. Define the absolute value of a real number x in this way:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0. \end{cases}$$

Give three examples showing that if x and y are real numbers, then $|xy| = |x| \cdot |y|$. Then, prove that this is true.

Solution.

We need to do two things: give **three examples** and then prove that

$$|xy| = |x||y|$$

for all real numbers x, y.

Three examples

Example 1: Let $x = 2$ and $y = 3$.

$$|xy| = |2(3)| = |6| = 6$$

and

$$|x||y| = |2||3| = 2(3) = 6.$$

Thus,

$$|xy| = |x||y|.$$

Example 2: Let $x = -4$ and $y = 5$.

$$|xy| = |-4(5)| = |-20| = 20$$

and

$$|x||y| = |-4||5| = 4(5) = 20.$$

Example 3: Let $x = -3$ and $y = -2$.

$$|xy| = |(-3)(-2)| = |6| = 6$$

and

$$|x||y| = |-3||-2| = 3(2) = 6.$$

So all three examples support the statement.

Proof

We consider the possible signs of x and y.

Case 1: $x \geq 0$ and $y \geq 0$

Then $xy \geq 0$. By the definition of absolute value,

$$|xy| = xy.$$

Also,

$$|x||y| = xy.$$

Therefore,

$$|xy| = |x||y|.$$

Case 2: $x \geq 0$ and $y < 0$

Then $xy < 0$, so

$$|xy| = -xy.$$

Since $x \geq 0$ and $y < 0$,

$$|x| = x, \qquad |y| = -y.$$

Thus,

$$|x||y| = x(-y) = -xy.$$

Therefore,

$$|xy| = |x||y|.$$

Case 3: $x < 0$ and $y \geq 0$

Then $xy < 0$, so

$$|xy| = -xy.$$

Also,

$$|x| = -x, \qquad |y| = y.$$

Therefore,

$$|x||y| = (-x)y = -xy.$$

Hence,

$$|xy| = |x||y|.$$

Case 4: $x < 0$ and $y < 0$

Then $xy > 0$, so

$$|xy| = xy.$$

Also,

$$|x| = -x, \qquad |y| = -y.$$

Therefore,

$$|x||y| = (-x)(-y) = xy.$$

Thus,

$$|xy| = |x||y|.$$

Since these four cases cover **all possible real values of x and y**, we conclude that

$$\boxed{|xy| = |x||y|}$$

for all real numbers x and y.